

A Preparation of Herbal Medicine *Salvia miltiorrhiza* Reduces Expression of Intercellular Adhesion Molecule-1 and Development of Atherosclerosis in Apolipoprotein E-Deficient Mice

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Abstract: Cardiogenic pill (CP) is a pharmaceutical preparation of the herbal medicine *Salvia miltiorrhiza*. In vitro studies demonstrate that CP inhibits vascular endothelial expression of adhesion molecules and smooth-muscle proliferation, implying the possibility of antiatherosclerotic effects. This study employs an in vivo animal model to examine the potential therapeutic efficacy of CP on atherosclerotic development. Male apolipoprotein E-deficient (ApoE^{-/-}) mice fed with an atherogenic (high fat) diet were administered with CP (90–120 mg/kg per day) via drinking water for 8 weeks. Hypercholesterolemia developed in the mice, with 22-fold increases in plasma levels of total cholesterol, 29-fold of LDL, and 7-fold of HDL. CP therapy did not significantly alter the lipid levels. Expression of intercellular adhesion molecule 1 significantly increased in circulating leukocytes and was abolished by CP therapy. Atherosclerosis significantly developed in the aorta and was attenuated by CP therapy, with an approximately 30% reduction in whole atherosclerotic lesions and an approximately 50% reduction in fibrous plaques in the artery. Thus, herbal medicine CP partly protects ApoE^{-/-} mice from high-fat diet-induced atherogenesis. The protection is unlikely to be attributable to decreases in circulating cholesterol levels, but it might possibly relate to an inhibition of expression of adhesion molecules and other effects that remain unknown at this time.

Key Words: atherosclerosis, apolipoprotein E-deficient mouse, hyperlipidemia, intercellular adhesion molecule 1, *Salvia miltiorrhiza* (*J Cardiovasc Pharmacol*TM 2008;51:38–44)

INTRODUCTION

Salvia miltiorrhiza (SM), named *Danshen* in Chinese, is a traditional herbal medicine commonly used in traditional Chinese medicine practice for the prevention and management of cardiovascular disease (CVD). Laboratory studies have demonstrated that SM possesses several beneficial actions on the cardiovascular system, including inhibition of cellular adhesion molecule release,^{1–5} reduction of LDL oxidation,^{6,7} and antiplatelet aggregation.⁸ The *cardiogenic pill* (CP), also named the *dantonic pill*, is a pharmaceutical agent in which SM and other two traditional Chinese medicines, *Panax notoginseng* and *borneol*, have been combined in a fixed ratio using modern techniques of pharmaceutical preparation according to the standard of good manufacturing practice. It is thought that the main ingredients in CP are hydrophilic phenolic acids and lipophilic tanshinones from SM. Although CP has recently been widely used in Chinese hospitals for the prevention and management of CVD, the physiological actions underlying these possible therapeutic effects are still not completely understood. Our previous in vitro study⁹ has demonstrated that CP can inhibit tumor necrosis factor α (TNF α)-induced expression of intercellular adhesion molecule 1 (ICAM-1) and vascular cell adhesion molecule 1 (VCAM-1) in vascular endothelial cells and that it can attenuate platelet-derived growth factor (PDGF)-induced proliferation of vascular smooth-muscle cells, suggesting a possible antiatherosclerotic action by this medicine. In the present study, the potential therapeutic effect of CP on atherosclerotic development has been examined through an in vivo animal model.

MATERIALS AND METHODS

Animals and Animal Husbandry

Male, 10-week-old, homozygous, apolipoprotein E-deficient (ApoE^{-/-}) mice were provided by the AMREP animal center (Melbourne, Australia) and maintained in accordance with guidelines of the AMREP animal ethics committee on the care and use of animals for laboratory research. The mice were randomly allocated to four groups of nine animals each, were housed in an aseptic environment under controlled conditions of light and humidity, and received food and water *ad libitum*.

For the study, animals were fed with normal laboratory mouse chow (normal diet) containing 4.3% fat and 0.02%

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cholesterol, or atherogenic chow (high-fat diet) containing 16% fat and 1% cholesterol (Glen Forrest Stockfeeders, Western Australia), with CP or placebo treatment, for 8 weeks. Animals were observed daily, and the intake of water and food was determined every other day; body weight was determined once per week. At the end of the experiment, the mice were killed by cervical dislocation, and the blood, heart, aorta, liver, lung, and kidney were collected for the laboratory assessment.

CP Preparation and Administration

Ethanol extract of *SM* and its pharmaceutical preparation, CP, were prepared in Tasly Pharmaceutical Co., Ltd. (Tianjin, China) with a standard procedure for the production of the CP. The extract is a brown, dry powder, in which amount of danshensu (3,4-dihydroxyphenyl lactic acid), a major bioactive compound in *SM*, is more than 3.5%, according to the quality standard. One capsule of CP (Lot No. 200304107) contains 0.25 g of *SM* extract, and one capsule of placebo (Lot No. 200304106) contains the same amount of starch. The therapeutic dosage of CP for clinical application is 3 g of the extract (12 capsules of CP) daily, or 50 to 60 mg of the extract per kilogram of body weight per day. In this study, 16 capsules of CP (4 g of the extract) or placebo were added into 1 L of the drinking water, which was freshly prepared twice per week, for free consumption by the animals, by which the medicine was consistently administered to the animals at high therapeutic concentrations (90–120 mg/kg per day) during the 8-week experimental period.

Identification of Bioactive Chemicals in CP

Four major bioactive compounds of *SM*, danshensu, protocatechuic aldehyde, salvanolic acid B, and tanshinone II-A (Fig. 1) were identified in the CP preparation, using an HPLC method.

All chemicals used were from Sigma Co. (St. Louis, MO) at HPLC analytical grade, except those specially indicated. Standard chemical danshensu was provided by the Tasly Pharmaceutical Co., and the chemicals protocatechuic aldehyde, salvanolic acid B, and tanshinone II-A sulfonate were purchased from the National Institute for Control of Biological and Pharmaceutical Products (Beijing, China). The standard chemicals were prepared in a 50% aqueous methanol stock

solution, and CP was dissolved in Milli-Q water and filtered through a 0.45- μ m membrane shortly before the analysis.

A Hewlett Packard 1090 Series II HPLC system and MZ Analysentechnik C-18 columns (300 Å, 5 μ m, 250 $\times$ 1.6 mm) were used for the analysis. The detection wavelength was set at 254 nm, and flow rate was 0.1 mL/min. The mobile phase was gradient elution mixed with solvent A [0.1% trifluoroacetic acid (TFA)] and B (acetonitrile with 0.1% TFA) degassed with ultrapure helium gas (Boc Gases Australia Ltd.). The gradient program was as follows: initial, minutes 0 to 20, linear change of A–B from 98:2 (v/v) to 77:23; minutes 20 to 35, 71.5:28.5; minutes 35 to 39, 20:80; and minutes 39 to 40, back to 98:2. Peaks in the chromatograms were identified by comparing the retention time and mobile phase with those of the standards.

Determination of Plasma Lipids and Glucose

Plasma was isolated from the whole-blood sample by centrifugation at 1500 rpm, 4°C, for 10 minutes, and levels of total cholesterol (TC), LDL, HDL, triglyceride (TG), and glucose were determined using commercially available enzymatic kits, following the manufacturer's instructions, performed regularly at the Alfred Hospital (Melbourne, Australia).

Analysis of Adhesion Molecule Expression

Expression of ICAM-1, VCAM-1, and E-selectin in circulating leukocytes was determined by fluorescence-activated cell sorting (FACS) analysis. Whole-blood samples were centrifuged at 1500 rpm, 4°C, for 10 minutes, and the centrifugal layer of white blood cells (buffy coat) was collected. The leukocytes were incubated in the lysis buffer for 10 minutes at 37°C and washed in PBS to remove red blood cells that possibly had mixed, and they were then resuspended in 0.5 mL of Hanks' balanced salt solution. FACS analysis was performed, using methods reported previously.⁹ In brief, cells were incubated with a specific fluorescence-connected antibody (Santa Cruz Biotechnology, CA) against ICAM-1, VCAM-1, or E-selectin for 1 hour; protected from light; then washed in 0.5 mL of Hanks' balanced salt solution to remove residual antibody, and then analyzed by the FACS scanning flow cytometer and CellQuest software (Becton Dickinson, Franklin Lakes, NJ).

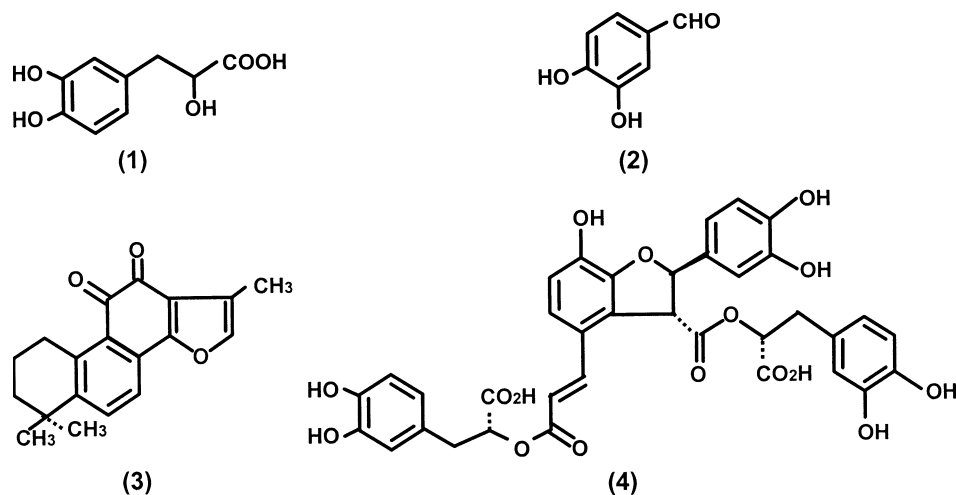


FIGURE 1. Chemical structures of four major components in *SM*: (1) danshensu; (2) protocatechuic aldehyde; (3) tanshinone II-A; and (4) salvanolic acid B.

Assessment of Atherosclerosis

Atherosclerosis formed mainly in the thoracic segment of the aorta in this animal model, and quantification of atherosclerotic lesions was performed by a method reported previously,¹⁰ with some modification. In brief, the thoracic aorta was isolated, embedded in OCT compound, and frozen in liquid nitrogen. Sequential, 5- and 10- μ m-thick cryosections were collected (about 40 sections for each sample) and mounted on poly-d-lysine-coated slides. Sections were stained with Oil Red O (10- μ m-thick sections) or hematoxylin and eosin (5- μ m-thick sections), respectively. The lesion areas, including those of fatty streak, fibrofatty, and fibrous plaque, were counted by light microscope. The ratios (percentage) of atherosclerotic lesion area to total surface area of the thoracic aorta and atherosclerotic plaque size to total lumen size of the aortic arch were calculated, respectively.

Histological Examination of Major Organs

The heart, liver, lung, and kidney were isolated and directly embedded in OCT or fixed in 40% formaldehyde and embedded in paraffin. Frozen OCT sections at 10 μ m and paraffin sections at 5 μ m were prepared, stained with Oil Red O and hematoxylin and eosin, respectively, and examined under a light microscope.

Statistical Analysis

All data were presented as means $\pm$ SEM. Comparison between data of two treatments was made by Student's *t* test, and comparison between three or more treatments, such as different time points by using two-way ANOVA, with post hoc testing by the Student–Neumann–Keul test. Differences at $P < 0.05$ were considered significant.

RESULTS

Existence of Main Bioactive Components of SM in the CP Preparation

The typical HPLC profile of danshensu, protocatechuic aldehyde, salvianolic acid B, and tanshinone II-A was completely separated (Fig. 2A), with appearances of danshensu at about 6.5 minutes (A–B: 92.8:8.7), protocatechuic aldehyde at about 9.5 minutes (88:12), tanshinone II-A at about 17 minutes (80.3:19.7), and salvianolic acid B at about 26.5 minutes (74.6:25.4), respectively. The four compounds were clearly identified as the major components in the CP preparation (Fig. 2B). Quantitative analysis was not successfully performed in the HPLC study, and the exact concentration of these chemicals in the CP was unknown. The HPLC profile of CP was similar to that of the SM extract (Fig. 2C), indicating that SM is the main ingredient in this herbal preparation.

Body Weight, Intakes of Food, and Water

Body weight increased more in the mice fed with normal diet (increase of 6.2 g per mouse, on average) than in those with high-fat diet (2.1 g per mouse) at the end of study. CP therapy did not significantly affect body weight in the mice fed with normal diet; in the high-fat-fed mice, average body weight seemed to be slightly increased during the CP

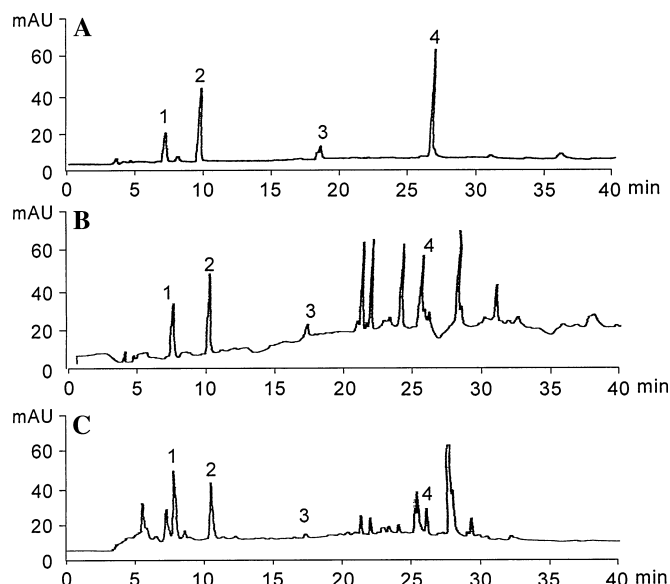


FIGURE 2. Representative HPLC chromatograms of (A) standard chemicals, (B) the CP preparation, and (C) SM extract. 1, danshensu; 2, protocatechuic aldehyde; 3, tanshinone II-A; 4, salvianolic acid B.

treatment, but at the end of study no statistical difference was found between the CP- and placebo-treated groups (Fig. 3). A significant reduction of daily food intake occurred in mice fed the high-fat diet (2.5 ± 0.4 g/d per mouse) compared with those with normal diet (3.8 ± 0.1 g/d per mouse, $P < 0.05$); daily water intake was also less in mice with the high-fat than the normal diet (2.6 ± 0.12 mL/d to 4.3 ± 0.3 mL/d per mouse, $P < 0.05$); CP therapy did not significantly affect the food and water intakes in all mice, except for slightly increased water intake in normal diet-fed mice, in comparison with the placebo treatment (Fig. 4).

Plasma Levels of Lipids and Glucose

Plasma levels of TC, LDL, and HDL, but not TG, significantly increased in the ApoE $^{-/-}$ mice at the end of 8-week experimental periods, with about 3- and 22-fold increases of TC, 3- and 29-fold increases of LDL, and 2- and 7-fold increases of HDL in the normal-diet- and atherogenic-diet-fed groups, respectively, in comparison with the wild-type mice (Table 1). CP treatment did not significantly alter the levels of the lipids. Plasma levels of glucose in the mice were not significantly changed by feeding with normal or high-fat diets, or treatment with the CP or placebo.

Attenuation of Atherogenic Diet-Induced Expression of ICAM-1 in Leukocytes by CP Therapy

The atherogenic diet significantly induced leukocyte expression of the adhesion molecule ICAM-1 in ApoE $^{-/-}$ mice, by an approximately sixfold increment of expression-positive cells in comparison with the normal diet, and CP treatment significantly abolished this increment of ICAM-1 expression (Fig. 5). There was a detectable expression of

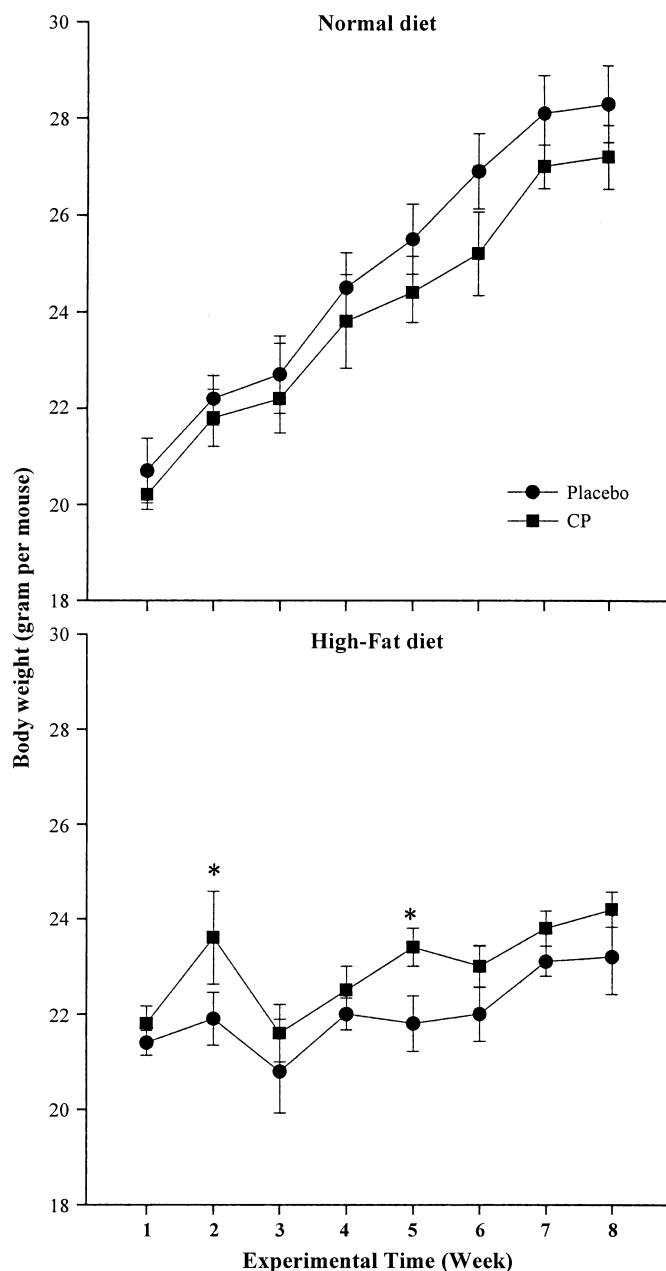


FIGURE 3. Body weight of ApoE^{-/-} mice fed normal or high-fat diets and treated with placebo or CP for 8 weeks. Data are shown as means $\pm$ SEM. $n = 9$, * $P < 0.05$ versus placebo-treated group at the same time point.

VCAM-1, but not E-selectin, in the circulating leukocytes. The rate of VCAM-1 expression was relatively much lower (~5%) and was not changed by either high-fat diet or CP treatment (data not shown).

Reduction of Atherogenic Diet-Induced Atherosclerotic Development by CP Therapy

In the ApoE^{-/-} mice fed with a normal diet, early-stage atherosclerosis, pathologically characterized as fatty streaks and mild fibrofatty lesions, could be identified in the

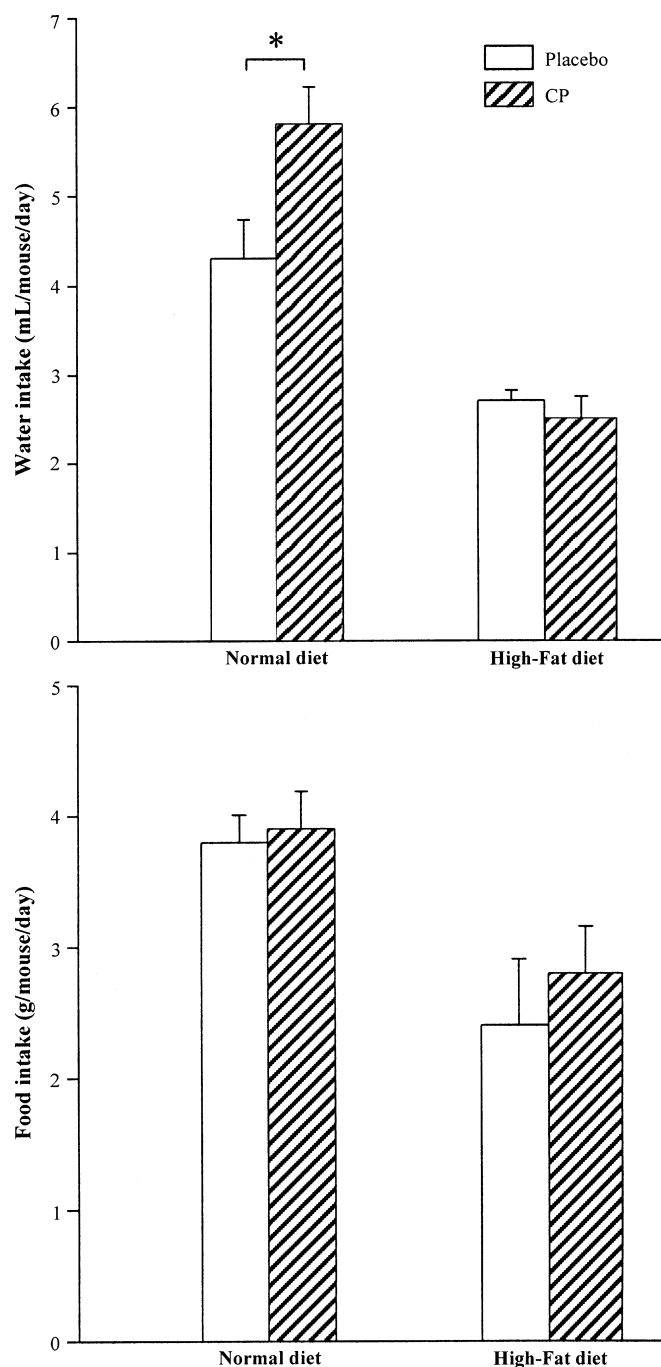


FIGURE 4. Intakes of water and food by ApoE^{-/-} mice fed normal or high-fat diets and treated with placebo or CP. Data are shown as means $\pm$ SEM of the average per day of 56 days (8 weeks), with nine mice for each group. * $P < 0.05$.

aorta, particularly in the proximal aorta, with the lesion area less than 5% in the whole aorta. The high-fat diet significantly accelerated atherosclerotic development. In the mice with an atherogenic diet for 8 weeks, typical atherosclerotic lesions, including fatty streaks, fibrofatty lesions, and fibrous plaques, with lesion areas consistently greater than 20%, were observed in the aorta, particularly in the thoracic segment of the artery,

TABLE 1. Levels of Plasma Lipids and Glucose in ApoE^{-/-} Mice Fed With Normal or High-Fat Diet and Treated With CP or Placebo

Group	TC	LDL	HDL	TG	Glucose
WT	3.52 ± 0.47	2.46 ± 0.32	1.39 ± 0.18	1.23 ± 0.10	7.0 ± 0.10
ND	11.40 ± 0.82*	7.67 ± 1.05*	3.13 ± 0.76*	1.33 ± 0.23	10.0 ± 0.85
ND + CP	10.93 ± 0.74*	7.24 ± 1.49*	3.65 ± 0.93*	1.50 ± 0.17	10.4 ± 0.38
HF	76.52 ± 3.97†	70.66 ± 3.90†	9.98 ± 1.88†	0.95 ± 0.10	9.27 ± 0.52
HF + CP	84.80 ± 5.40†	67.11 ± 3.90†	8.20 ± 0.87†	0.95 ± 0.13	7.30 ± 0.50

Mean ± SEM (mM), n = 9.

WT, wild-type mice; ND, normal diet; HF: high-fat diet.

**P* < 0.01 versus WT.†*P* < 0.01 versus WT and ND.

with Oil Red O plus hemotoxylin staining. CP treatment significantly reduced the high-fat-diet-induced atherogenesis: the area of atherosclerotic lesions was $18.8 \pm 2.5\%$ in the whole thoracic aorta of CP-treated mice and $26.7 \pm 3.6\%$ in the mice that had received the placebo treatment (*P* < 0.05) (Fig. 6A), and the size of atherosclerotic plaques in the aortic arch was significantly less in CP- ($8.5 \pm 1.3\%$) than in placebo-treated mice ($18.3 \pm 3.5\%$, *P* < 0.01) (Fig. 6B).

Histological Changes in Other Organs

In ApoE^{-/-} mice fed a normal diet, a moderate degree of lipidosis was found in the liver and lungs, but not in the heart and kidneys; the high-fat diet caused a significant increase in lipidosis in the liver and lungs, a mild lipidosis in renal glomeruli, and no significant histological changes in the heart; there was no difference between the CP and placebo

treatment in histological changes in the heart, liver, lungs, and kidneys (data not shown).

DISCUSSION

The present study shows that treatment with the herbal medicine *SM* in the form of the CP preparation significantly reduces the progression of aortic atherosclerosis in ApoE-deficient mice fed with a high-fat diet, as indicated by histological assessment of the atherosclerotic area and fibrous plaque size in the artery. The antiatherosclerotic effect was paralleled by a marked decrease in adhesion molecule ICAM-1 expression in circulating leukocytes, but not by changes in plasma cholesterol levels. Four important bioactive chemical compounds of *SM*—danshensu, protocatechuic aldehyde, salvianolic acid B, and tanshinone II-A—were identified in the CP preparation by HPLC

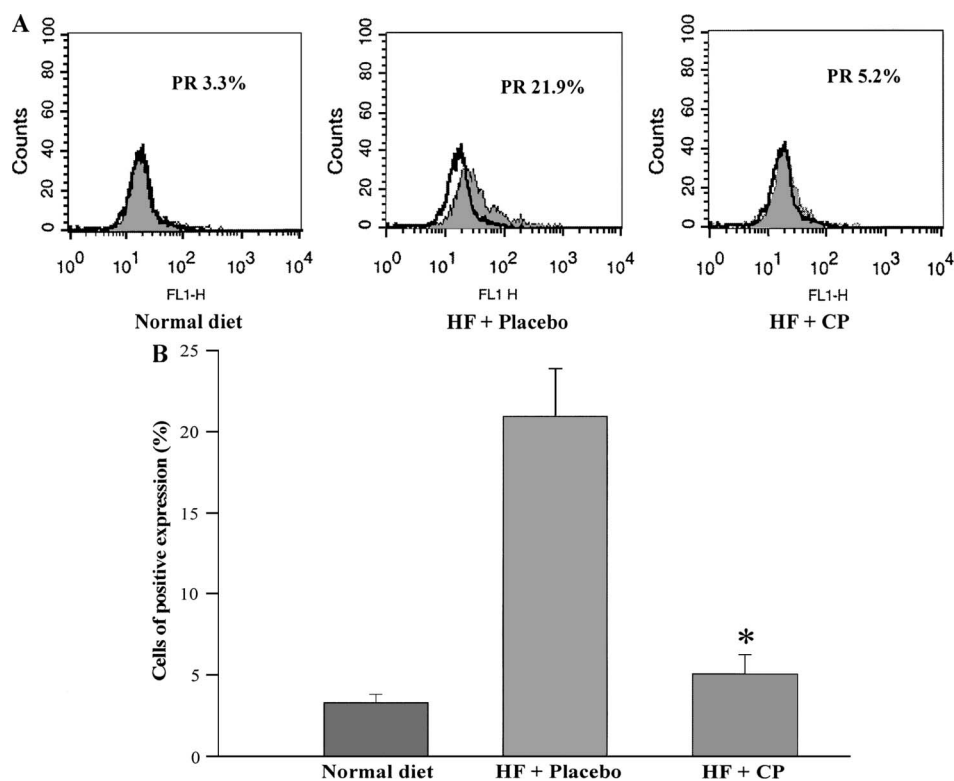


FIGURE 5. FACS scanning analysis of ICAM-1 expression in circulating leukocytes of ApoE^{-/-} mice fed normal or high-fat (HF) diets and treated with placebo or CP for 8 weeks. **A**, Representative FACS histograms; the shaded area indicates the cells from each group, and the bold black line indicates the sorting of negative cells; PR, positive rate. **B**, Average percentages (mean ± SEM) of positive cells in different treated groups. n = 9, **P* < 0.01 versus the placebo-treated group.

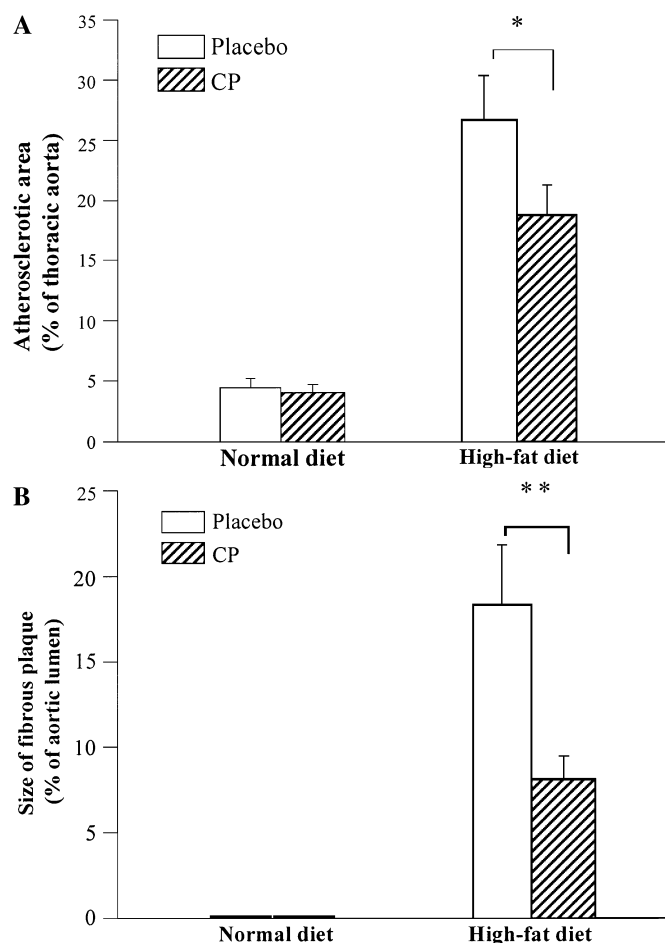


FIGURE 6. Quantitation of atherosclerotic lesions in the thoracic aorta of ApoE^{-/-} mice fed normal or high-fat diets and treated with placebo or CP for 8 weeks by light microscopy with Oil Red O and hematoxylin and eosin stain. Data are shown as means $\pm$ SEM. $n = 9$, $*P < 0.05$, $**P < 0.01$. A, The average percentage of the area of atherosclerotic lesions in the whole thoracic aorta in different groups. B, The average percentage of fibrous plaque size in the whole aortic arch in different groups.

analysis. Therefore, the present results support the concept that CP may be an effective antiatherosclerotic agent and that its modes of action on the atherosclerotic processes are related to the known cardiovascular benefits of the herbal medicine *SM*, such as reduction of expression of adhesion molecules and antiinflammatory properties.¹¹

Atherosclerosis is the primary pathological lesion responsible for most (>90%) clinical cardiovascular disease. Although hypercholesterolemia has traditionally been considered a major cause of atherogenesis, the role of inflammation in atherosclerosis has been increasingly recognized in recent years.¹² In response to atherogenic factors, mononuclear cells in blood attach and adhere to, and spread on, the luminal surface of the arterial tree, particularly at branches and bifurcations, initiating atherosclerotic formation.

Cellular adhesion molecules, including ICAM-1, VCAM-1, and E-selectin, which are expressed on the vascular

endothelium and circulating leukocytes in response to several inflammatory stimuli, mediate the adhesion of leukocytes to the vascular endothelium, the pivotal early event in atherogenesis.¹³ Increased expression of these molecules is observed in atherosclerotic lesions.¹⁴ Increased circulating levels of ICAM-1 and VCAM-1 are not only a biomarker, but also a potential therapeutic target, for atherosclerosis.¹³ Animal studies demonstrate that selective inhibition of these molecules reduces atherogenesis. For example, a monoclonal antibody against VCAM-1 profoundly reduces neointimal formation after carotid injury in genetically hypercholesterolemic mice¹⁵; antibody blockade of integrin $\alpha 4\beta 1$ (the ligand for VCAM-1) reduces intimal hyperplasia in endarterectomized carotid arteries in a primate model¹⁶; mice deficient in the adhesion molecules ICAM-1, VCAM-1, or P-selectin have reduced responses to experimental atherosclerotic stimuli and atherosclerotic lesions¹⁷; anti-CD40L treatment in hyperlipidemic mice results in reduced atherosclerosis through reduction of VCAM-1 expression and inhibition of inflammatory cell accumulation¹⁸; and inhibition of circulating level and vascular endothelial expression of ICAM-1 by a selective thromboxane receptor block S18886 relates to reduction of atherogenesis in ApoE-deficient mice.¹⁹

The inhibitory effect of herbal medicine *SM* on adhesion molecule expression has been demonstrated by several in vitro studies, and the main underlying chemical compounds for this effect are the water-soluble phenolic acids, particularly protocatechuic aldehyde and salvianolic acid B in the herb.¹⁻⁵ The CP, in which protocatechuic aldehyde and salvianolic acid B were identified to be the major components by HPLC analysis (Fig. 2), was shown to inhibit ICAM-1 and VCAM-1 expression in vascular endothelial cells in an early study.⁹ The present study has shown that CP therapy abolishes atherogenic, diet-induced ICAM-1 expression in circulating leukocytes in ApoE-deficient mice (Fig. 5), confirming the anti-ICAM-1 property of this herbal preparation for in vivo situations. In this study, the effects of CP on production of other kinds of adhesion molecules are still unclear. The adhesion molecules VCAM-1 and E-selectin generally distribute in the vascular endothelium. VCAM-1 may be expressed in other kinds of cells under disease conditions.^{20,21} Our study shows that the expression of VCAM-1 was much lower, and no expression of E-selectin was found in the circulating leukocytes from mice with either normal or high-fat diets. The study could not exclude a possible role of CP on VCAM-1 and E-selectin, because their expression in endothelium and soluble form in plasma were not assessed. These studies, taken together, support the possibility that reduction of expression of adhesion molecules is an important property of the CP. The antiadhesion molecule action, together with other possible actions such as antioxidation and inhibition of platelet aggregation,⁶⁻⁸ from *SM*, may contribute to the preventive effect of CP on the development of atherosclerosis.

The herbal medicine *SM* contains multiple chemicals. These can generally be divided into two categories: hydrophilic and lipophilic compounds. More than 50 hydrophilic and 30 lipophilic compounds have been separated and identified from this herb, among which hydrophilic phenolic acids—particularly danshensu, protocatechuic aldehyde, and

salvianolic acid—and lipophilic tanshinones, such as tanshinone IIA, are the most important components responsible for many biological activities of *SM*.²² In the present study, danshensu, protocatechuic aldehyde, salvianolic acid B, and tanshinone IIA were identified in the CP preparation as the major components. At this time, the blood levels of these compounds, and their secretion and metabolism by the body during CP therapy, as well as the specific action by each compound, are still unknown.

CONCLUSION

The herbal preparation CP protects ApoE-deficient mice from atherosclerosis induced by an atherogenic diet. This protection seems to be caused not by a reduction in circulating levels of cholesterol, but by other mechanisms, including reduction of adhesion molecule production and, perhaps, other unknown factors. Further studies are needed to examine the possible mechanisms underlying the inhibition of atherosclerosis by this *SM* preparation, and to identify the compounds responsible for this effect. Such studies should also identify side effects and assist in the development of procedures for ensuring quality control of this medicine, before consideration of large-scale clinical applications for the prevention and management of atherosclerotic cardiovascular diseases.

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